

TrES-5b

R-0866

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_240921 · created 2026-07-28T22:45:17+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460574.7975 +/- 0.0041 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.053
Transit depth (Rp/Rs)^2	3.12 +/- 1.87 [%]
Orbital Inclination (inc)	89.07 +/- 2.9
Airmass coefficient 1 (a1)	0.734 +/- 0.07
Airmass coefficient 2 (a2)	0.22 +/- 0.19
Scatter in the residuals of the lightcurve fit is	10.04 %
Best Comparison Star	#1 - [556, 81]
Optimal Aperture	2.31
Optimal Annulus	8.6
Transit Duration (day)	0.0863 +/- 0.007

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.177 $\pm$ 0.053 vs 0.142 (deviation 0.59 σ)
Duration fit vs published	0.0863 d vs 0.0758 d (deviation 1.34 σ)
Mid-time O-C	9.4 min
Depth SNR	3.0
Residual scatter	10.04 %

Light curve

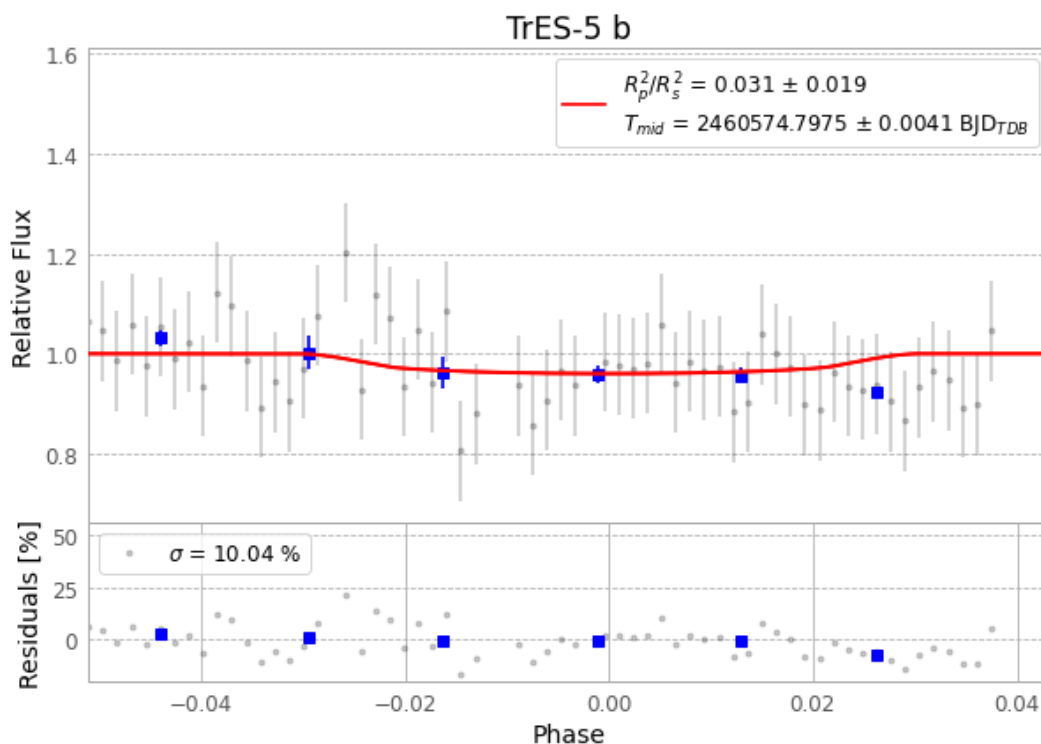


Figure 1 — FinalLightCurve_TrES-5 b_2024-09-20.png (SHA-256 c3c914e91e29c1a6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [355, 284], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 675a52dc40b39758...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

71 FITS frames (47 MB), TRES-5240921051516.FITS → TRES-5240921084515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-240921.log (521fda5890c0...), FinalLightCurve_TrES-5 b_2024-09-20.png (c3c914e91e29...), FinalLightCurve_TrES-5 b_2024-09-20.csv (fa71785873cb...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 22:45Z.